Subroutines

Parameters Passing

Methods of Passing Parameters

- In Registers
  Must document registers used
  Limited number of registers
  Uses fast on-chip registers
  Dictates register usage

- In a Parameter Block
  Parameters placed in a data structure
  Location of structure passed in a register
  Only uses one register
  Unlimited number of parameters
  Uses slow off-chip memory

More Parameter Passing Methods

- On the Stack
  Push parameters onto the stack
  Unlimited number of parameters
  Uses slow off-chip memory
  Don't have to manage memory
  Need a calling convention

	C/C++	Pascal
order parameters pushed onto stack	reverse order	given order
who clears stack	caller	subroutine
return value	in a register	on the stack

The Stack Frame

- Normally used by Compilers
- Creates a hole in the stack for private use by the subroutine (local variables)
- Requires a *base* register to point to the start of the frame (SP points to the end)
- Subroutine parameters kept above the frame
- Local's stored inside the frame
- Stack can still be used for temporary storage

Operation of a Stack Frame

Param $\mathcal{C}$		
Param $\mathcal{B}$		[R12, #8]
Param $\mathcal{A}$		
Local a	$\Leftarrow$ FP	
Local b		[R12, #-4]
Local c		
	$\Leftarrow$ SP	

1. Push parameters ($\mathcal{A}$, $\mathcal{B}$, and $\mathcal{C}$)
2. Reserve space for local variable (a, b, and c)
   Use R12 as Frame Pointer (FP)
   Stack can be used as normal
3. Return from subroutine / Clear Locals
4. Clear Stack / Pop parameters

Types of Parameter Passed

- Pass by value (most common)
  Value of the variable is used
- Pass by reference (common)
  Location (address) of variable is used
  Allows the subroutine to modify the original variable
- Pass by name (rare)
  A string holding the name of the data field is passed
  Subroutine uses string to access data (table or field)

Summary of Parameter Passing

- Methods of passing parameters
  Pass in Register
  Pass in Parameter Block
  Pass on Stack

- The Stack Frame
  Parameter Block on the Stack
  Used by Compilers

- Type of parameter passed
  Pass by value
  Pass by reference
  Pass by name